

TPS65219 PMIC EVM

Power-Rail Measurement Report

Generated : 07 May 2026 11:52:05
Instrument : Keysight EDUX1052A | Timebase : 200 ms/div | BUCK : tRAMP (10%→98%) | LDO : Slew (10%→90%)

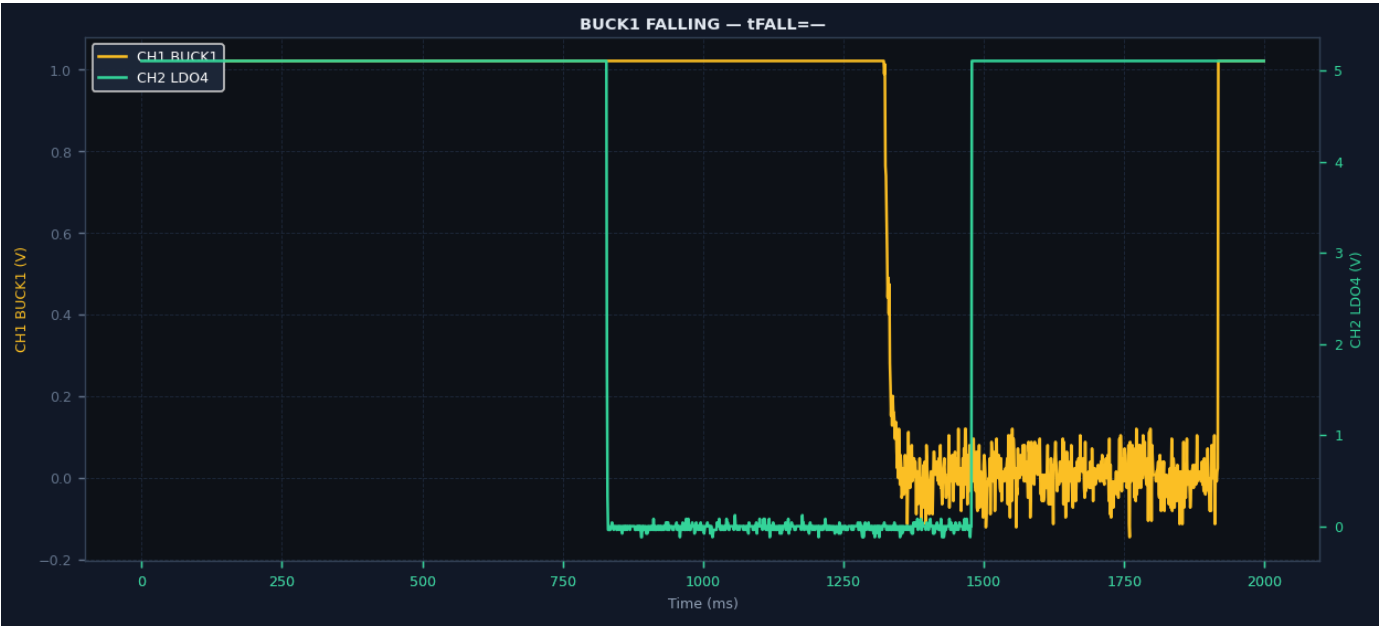
Total Events	Rails Tested	Snapshots
17	6	17

Rail Configuration

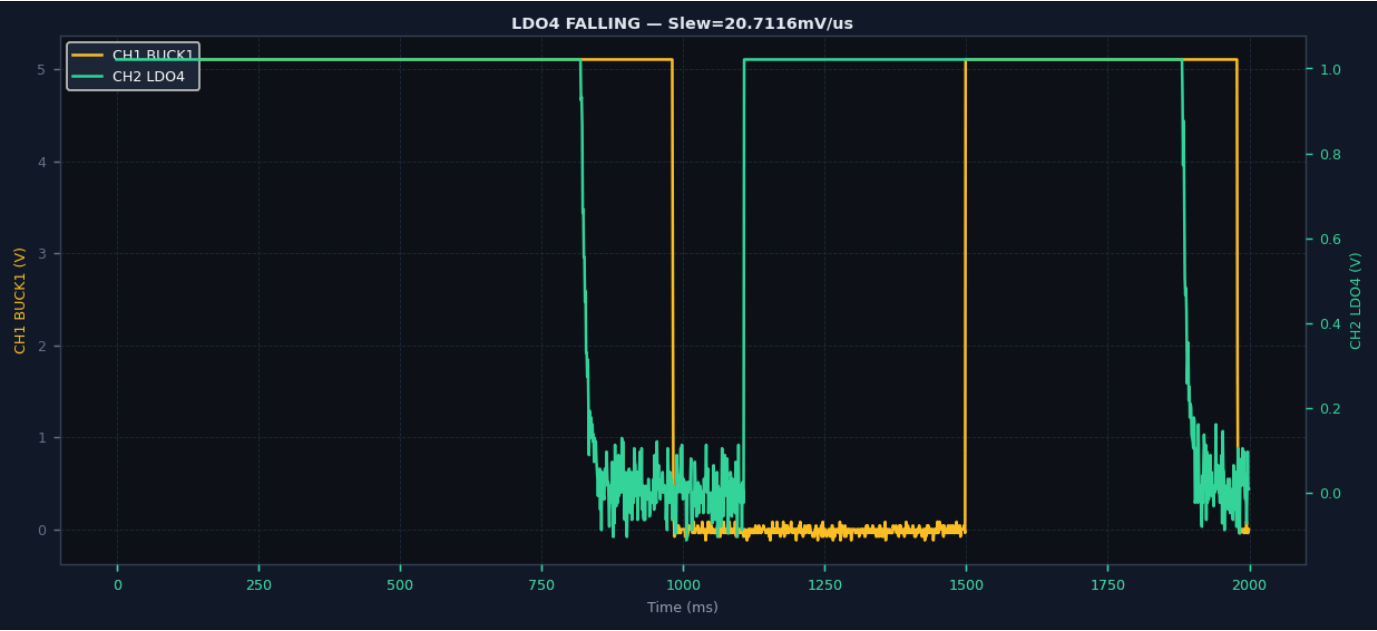
Rail	Type	VNOM (V)	V/div	Trig V	Spec Param	Spec Min	Spec Max	Unit
BUCK1	Buck DC-DC	0.749	0.20	0.10	ms tRAMP	1.0	500.0	ms
BUCK2	Buck DC-DC	3.289	1.00	0.30	ms tRAMP	1.0	500.0	ms
BUCK3	Buck DC-DC	1.197	0.50	0.10	ms tRAMP	1.0	500.0	ms
LDO1	LDO type 1	1.796	0.50	0.20	mV/μs Slew	10.0	14.0	mV/μs
LDO2	LDO type 1	0.848	0.20	0.10	mV/μs Slew	10.0	14.0	mV/μs
LDO3	LDO type 2	1.796	0.50	0.20	mV/μs Slew	7.5	27.0	mV/μs
LDO4	LDO type 2	2.496	1.00	0.20	mV/μs Slew	7.5	27.0	mV/μs

Waveform Snapshots & Results

Snapshot 1 | Rail: BUCK1 | CH: CH1 | Direction: FALLING | VNOM=0.749V | Result: —



Parameter	Measured	Spec/Reference	Result
Rail	BUCK1	VNOM=0.749V	--
Direction	FALLING	--	--
Method	98%→10% swing	--	--
Measurement	tFALL=—	—	—
DC Voltage	0.591V	-0.251–1.749V	PASS



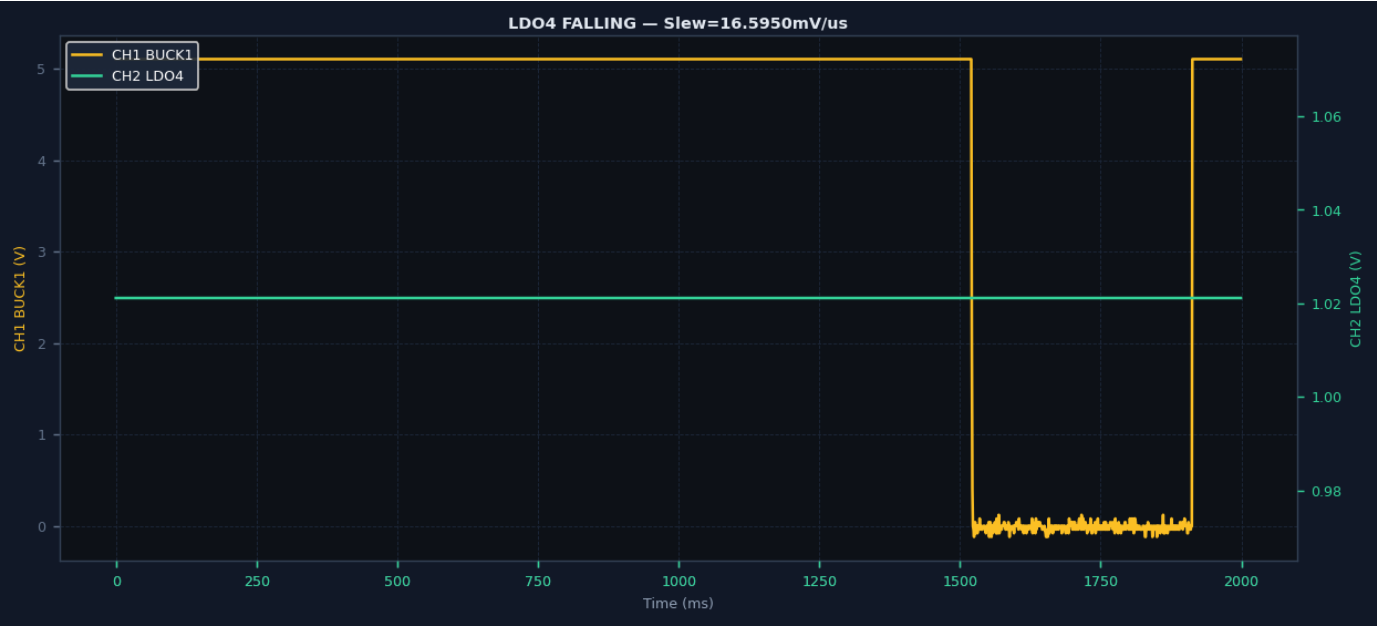
Parameter	Measured	Spec/Reference	Result
Rail	LDO4	VNOM=2.496V	--
Direction	FALLING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=20.7116mV/us	7.5–27.0mV/us	PASS
DC Voltage	2.941V	1.496–3.496V	PASS



Parameter	Measured	Spec/Reference	Result
Rail	BUCK1	VNOM=0.749V	--
Direction	RISING	--	--
Method	10%→98% swing	--	--
Measurement	tRAMP=—	1.0–500.0ms	—
DC Voltage	0.745V	-0.251–1.749V	PASS



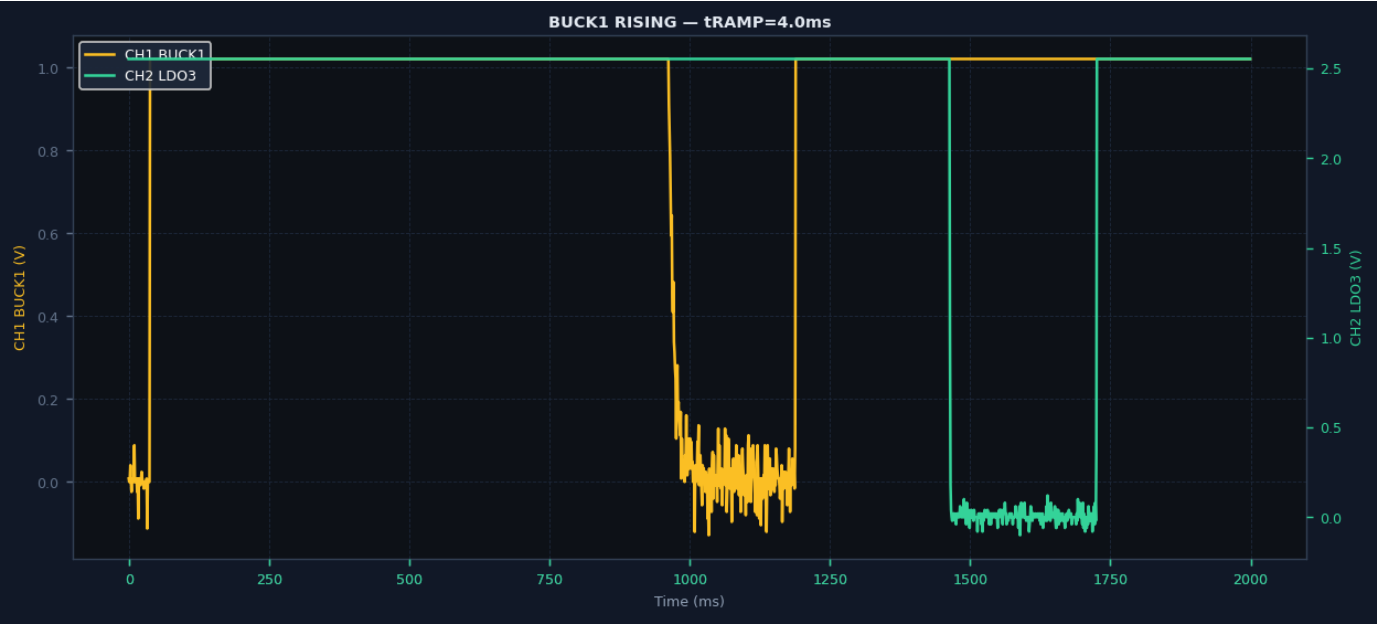
Parameter	Measured	Spec/Reference	Result
Rail	LDO4	VNOM=2.496V	--
Direction	RISING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=—	7.5–27.0mV/us	—
DC Voltage	5.106V	1.496–3.496V	PASS



Parameter	Measured	Spec/Reference	Result
Rail	LDO4	VNOM=2.496V	--
Direction	FALLING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=16.5950mV/us	7.5–27.0mV/us	PASS
DC Voltage	3.435V	1.496–3.496V	PASS



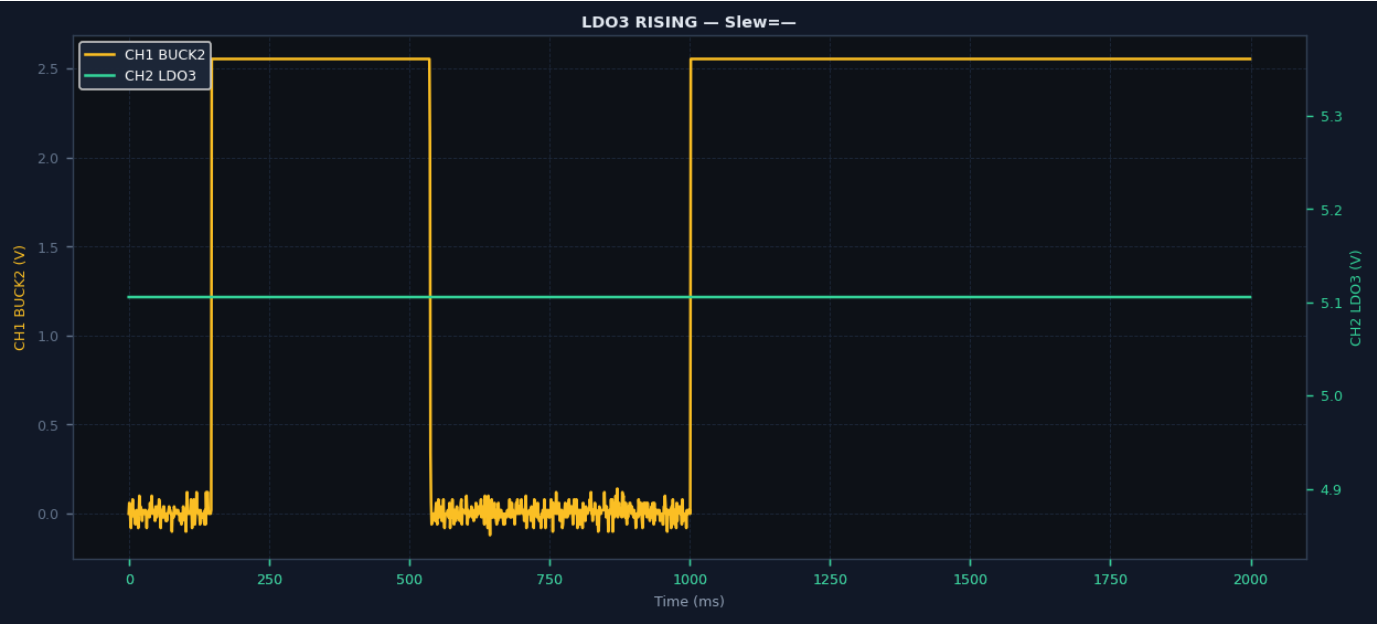
Parameter	Measured	Spec/Reference	Result
Rail	LDO3	VNOM=1.796V	--
Direction	RISING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=8.4004mV/us	7.5–27.0mV/us	PASS
DC Voltage	2.553V	0.796–2.796V	PASS



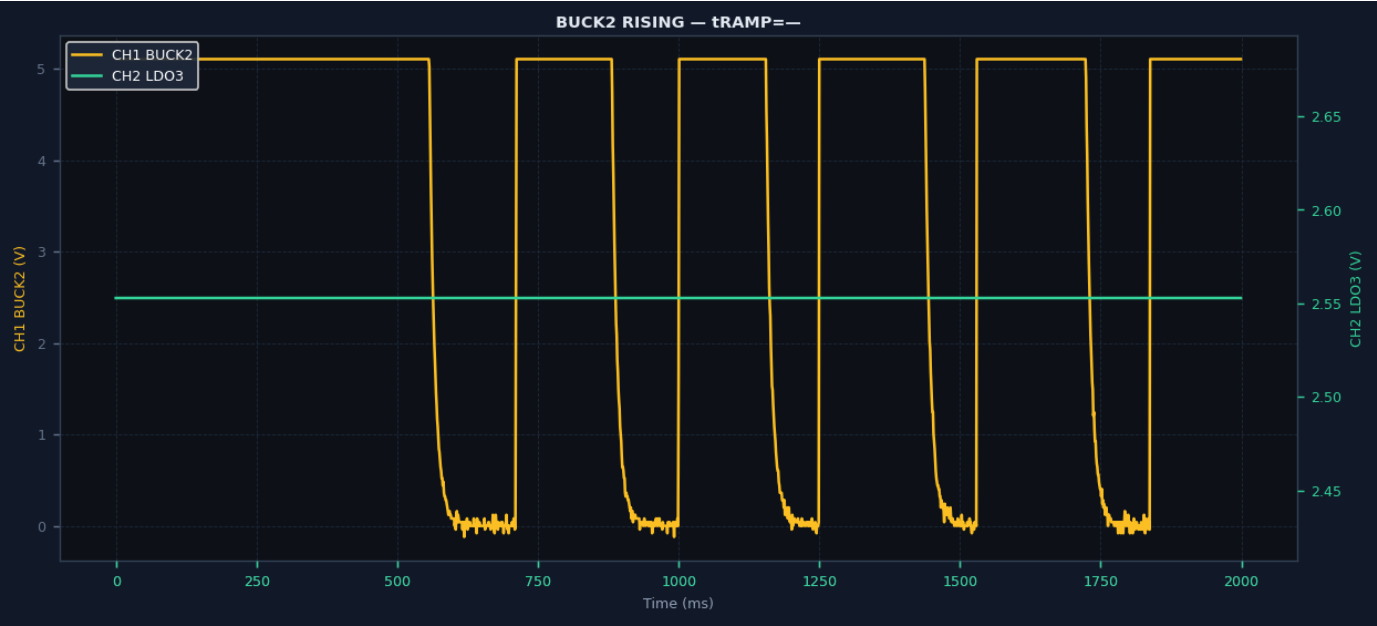
Parameter	Measured	Spec/Reference	Result
Rail	BUCK1	VNOM=0.749V	--
Direction	RISING	--	--
Method	10%→98% swing	--	--
Measurement	tRAMP=4.0ms	1.0–500.0ms	PASS
DC Voltage	0.982V	-0.251–1.749V	PASS



Parameter	Measured	Spec/Reference	Result
Rail	LDO3	VNOM=1.796V	--
Direction	FALLING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=—	7.5–27.0mV/us	—
DC Voltage	2.130V	0.796–2.796V	PASS



Parameter	Measured	Spec/Reference	Result
Rail	LDO3	VNOM=1.796V	--
Direction	RISING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=—	7.5–27.0mV/us	—
DC Voltage	2.393V	0.796–2.796V	PASS



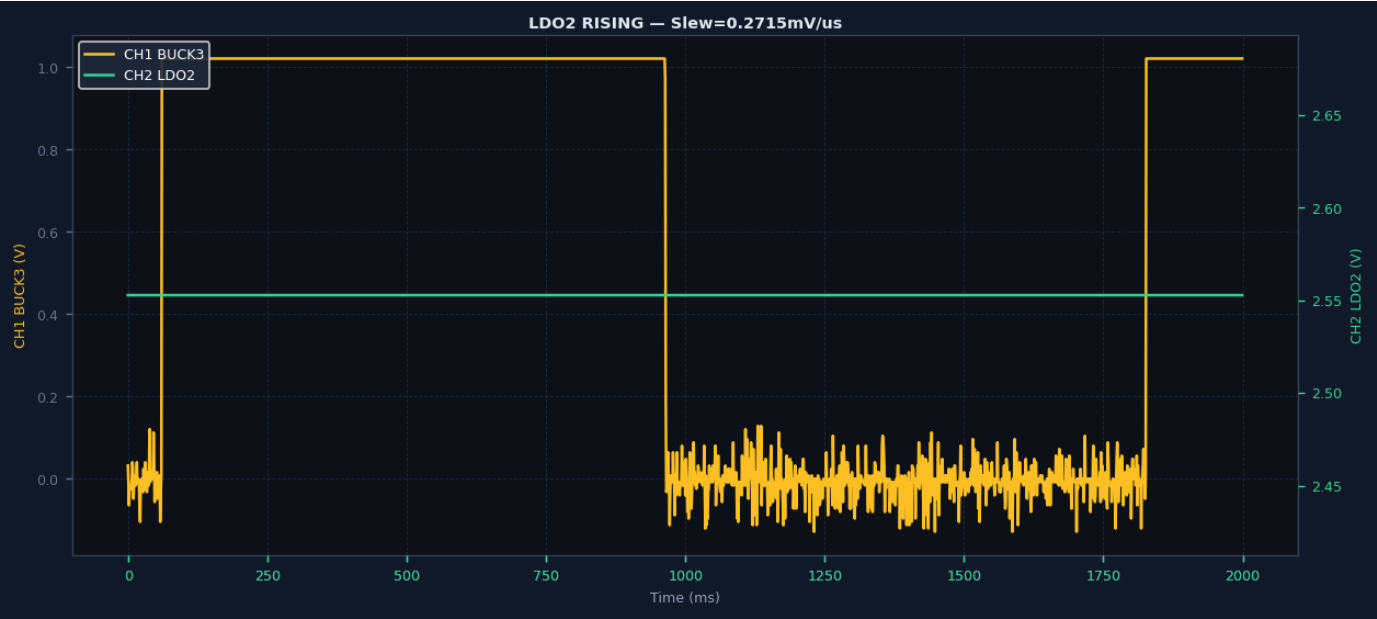
Parameter	Measured	Spec/Reference	Result
Rail	BUCK2	VNOM=3.289V	--
Direction	RISING	--	--
Method	10%→98% swing	--	--
Measurement	tRAMP=—	1.0–500.0ms	—
DC Voltage	5.106V	2.289–4.289V	PASS



Parameter	Measured	Spec/Reference	Result
Rail	BUCK2	VNOM=3.289V	--
Direction	FALLING	--	--
Method	98%→10% swing	--	--
Measurement	tFALL=27.0ms	—	PASS
DC Voltage	2.455V	2.289–4.289V	PASS



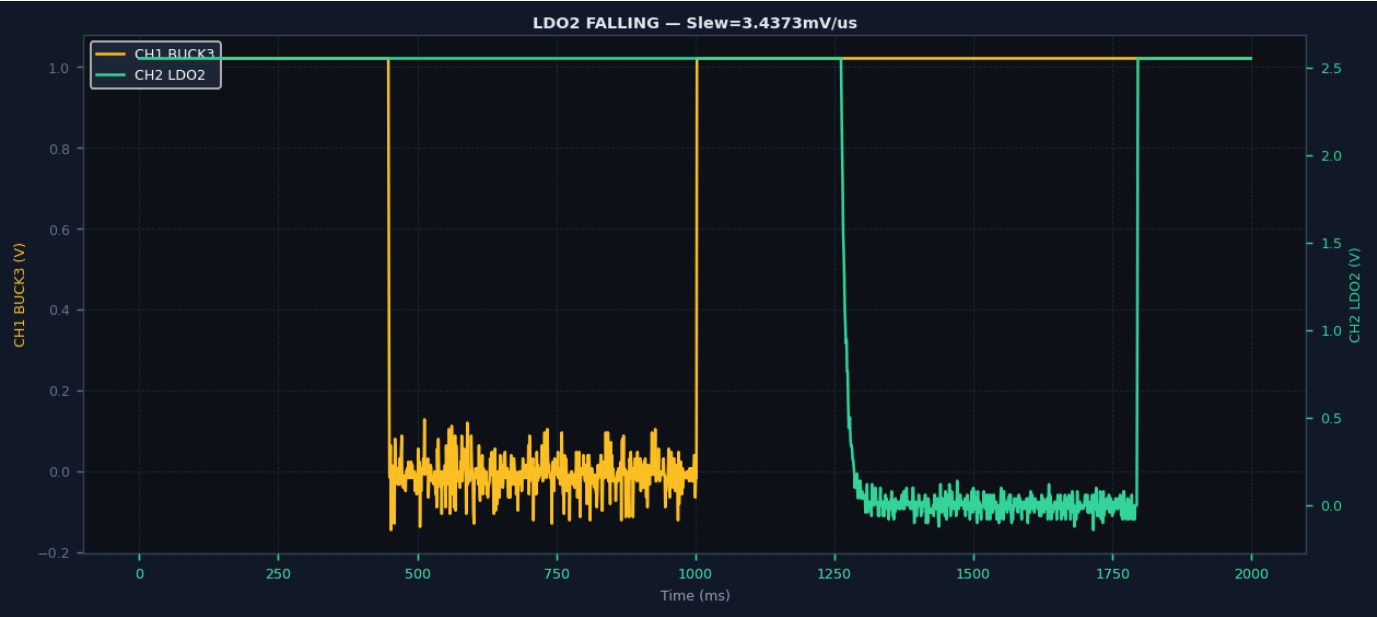
Parameter	Measured	Spec/Reference	Result
Rail	LDO3	VNOM=1.796V	--
Direction	FALLING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=8.3232mV/us	7.5–27.0mV/us	PASS
DC Voltage	2.553V	0.796–2.796V	PASS



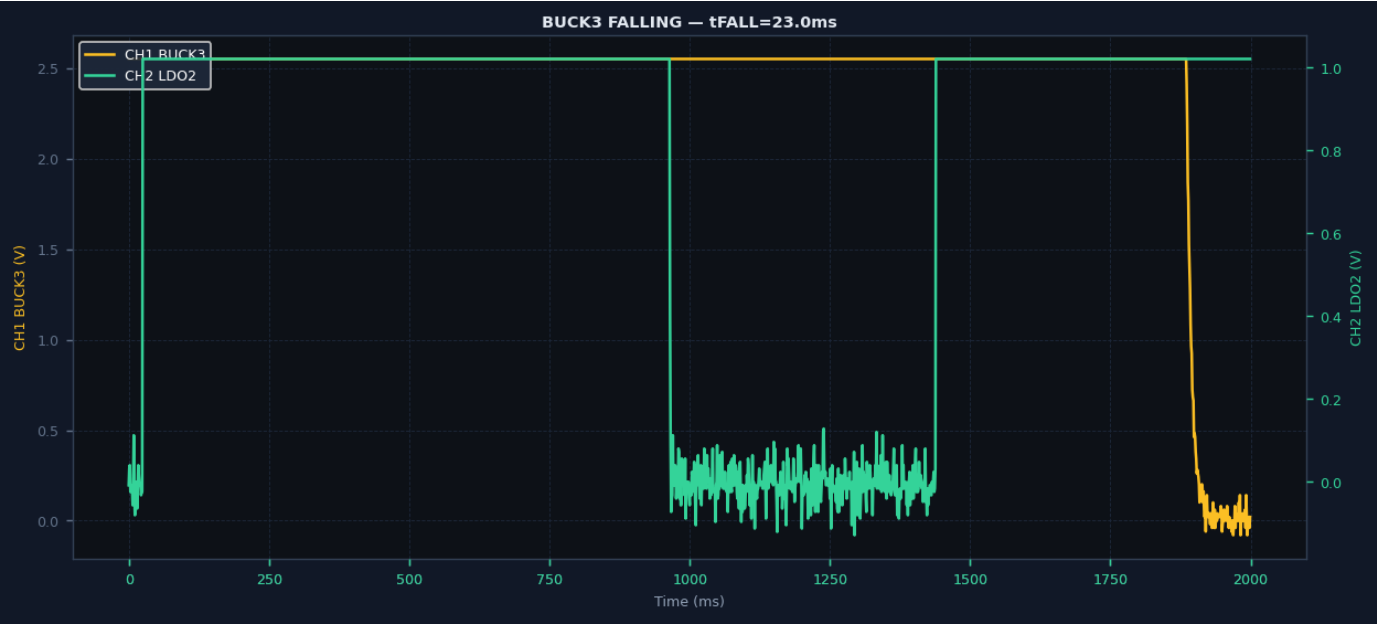
Parameter	Measured	Spec/Reference	Result
Rail	LDO2	VNOM=0.848V	--
Direction	RISING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=0.2715mV/us	10.0–14.0mV/us	FAIL
DC Voltage	0.503V	-0.152–1.848V	PASS



Parameter	Measured	Spec/Reference	Result
Rail	BUCK3	VNOM=1.197V	--
Direction	RISING	--	--
Method	10%→98% swing	--	--
Measurement	tRAMP=—	1.0–500.0ms	—
DC Voltage	2.154V	0.197–2.197V	PASS



Parameter	Measured	Spec/Reference	Result
Rail	LDO2	VNOM=0.848V	--
Direction	FALLING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=3.4373mV/us	10.0–14.0mV/us	PASS
DC Voltage	1.021V	-0.152–1.848V	PASS



Parameter	Measured	Spec/Reference	Result
Rail	BUCK3	VNOM=1.197V	--
Direction	FALLING	--	--
Method	98%→10% swing	--	--
Measurement	tFALL=23.0ms	—	PASS
DC Voltage	1.490V	0.197–2.197V	PASS



Parameter	Measured	Spec/Reference	Result
Rail	LDO2	VNOM=0.848V	--
Direction	RISING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=3.4425mV/us	10.0–14.0mV/us	PASS
DC Voltage	0.504V	-0.152–1.848V	PASS

Complete Measurement Log

#	Rail	Ch	Dir	Measured	Spec Range	DC (V)	P/F
1	BUCK1	CH1	FALLING	tFALL=—	—	0.591V	—
2	LDO4	CH2	FALLING	Slew=20.7116mV/us	7.5–27.0mV/us	2.941V	PASS
3	BUCK1	CH1	RISING	tRAMP=—	1.0–500.0ms	0.745V	—
4	LDO4	CH2	RISING	Slew=—	7.5–27.0mV/us	5.106V	—
5	LDO4	CH2	FALLING	Slew=16.5950mV/us	7.5–27.0mV/us	3.435V	PASS
6	LDO3	CH2	RISING	Slew=8.4004mV/us	7.5–27.0mV/us	2.553V	PASS
7	BUCK1	CH1	RISING	tRAMP=4.0ms	1.0–500.0ms	0.982V	PASS
8	LDO3	CH2	FALLING	Slew=—	7.5–27.0mV/us	2.130V	—
9	LDO3	CH2	RISING	Slew=—	7.5–27.0mV/us	2.393V	—
10	BUCK2	CH1	RISING	tRAMP=—	1.0–500.0ms	5.106V	—
11	BUCK2	CH1	FALLING	tFALL=27.0ms	—	2.455V	PASS
12	LDO3	CH2	FALLING	Slew=8.3232mV/us	7.5–27.0mV/us	2.553V	PASS
13	LDO2	CH2	RISING	Slew=0.2715mV/us	10.0–14.0mV/us	0.503V	FAIL
14	BUCK3	CH1	RISING	tRAMP=—	1.0–500.0ms	2.154V	—
15	LDO2	CH2	FALLING	Slew=3.4373mV/us	10.0–14.0mV/us	1.021V	PASS
16	BUCK3	CH1	FALLING	tFALL=23.0ms	—	1.490V	PASS
17	LDO2	CH2	RISING	Slew=3.4425mV/us	10.0–14.0mV/us	0.504V	PASS

Specification Reference

Rail	Parameter	Method	Min	Max	Unit
BUCK1/2/3	tRAMP	10%→98% swing	1.0	500.0	ms (EVM)
BUCK IC	tRAMP	IC datasheet	0.30	1.65	ms
LDO1/LDO2	Slew	10%→90% swing	10	14	mV/μs
LDO3/LDO4	Slew	10%→90% swing	7.5	27	mV/μs

DC voltage checked against nominal specification range. All waveform measurements at 200ms/div on Keysight EDUX1052A.